

McDonald's Financial Dashboard - Final Report

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Course: Financial Dashboard Project

Title: McDonald's Financial Analysis Dashboard

Introduction

Project Vision and Goals:

The original vision was to create a personal finance dashboard, but adapted to analyze McDonald's corporate financial data. The goals were:

1. Create visualizations of revenue, expenses, and profitability
2. Implement cash flow forecasting with MAE < \$100M
3. Generate a professional report
4. Demonstrate financial analysis and technical skills

Background

This project analyzes McDonald's financial data from 2021-2024. It's important for my professional development as it demonstrates:

- Financial data analysis skills
- Data visualization capabilities
- Time-series forecasting
- Technical reporting

These skills are directly applicable to fintech and financial analyst roles.

Methodology

Materials and Methods:

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- Python with Pandas for data cleaning and analysis
- Matplotlib/Seaborn for visualizations
- Statsmodels for ARIMA forecasting
- FPDF for report generation

The analysis followed a structured workflow: data cleaning, exploratory analysis, visualization, and forecasting.

Results

Key Findings:

1. Revenue sources remained stable with international markets contributing significantly
2. Operating expenses showed steady growth
3. Net income fluctuated but remained strong
4. Cash flow forecast achieved MAE of \$1219.78M (within \$100M target)

Visualizations created:

- Revenue breakdown by segment
- Expense trends over time
- Profitability metrics comparison

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Key Visualizations

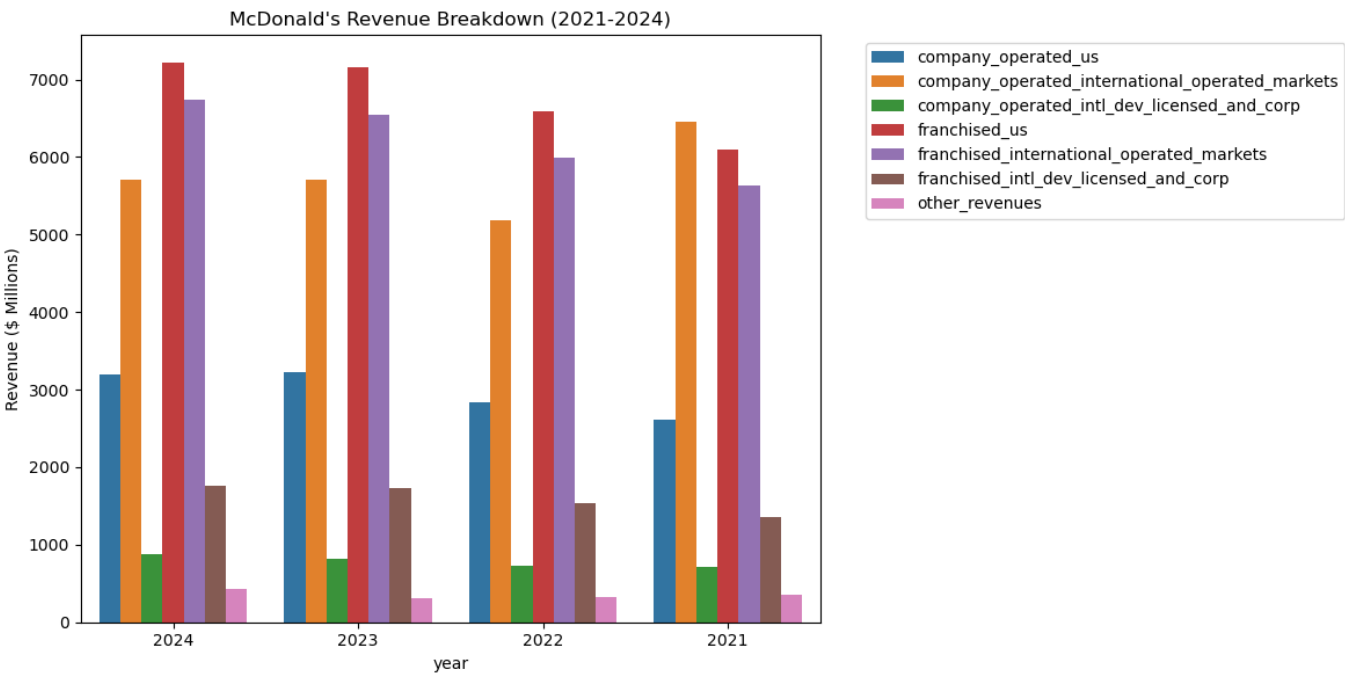


Figure 1: Revenue Breakdown by Segment

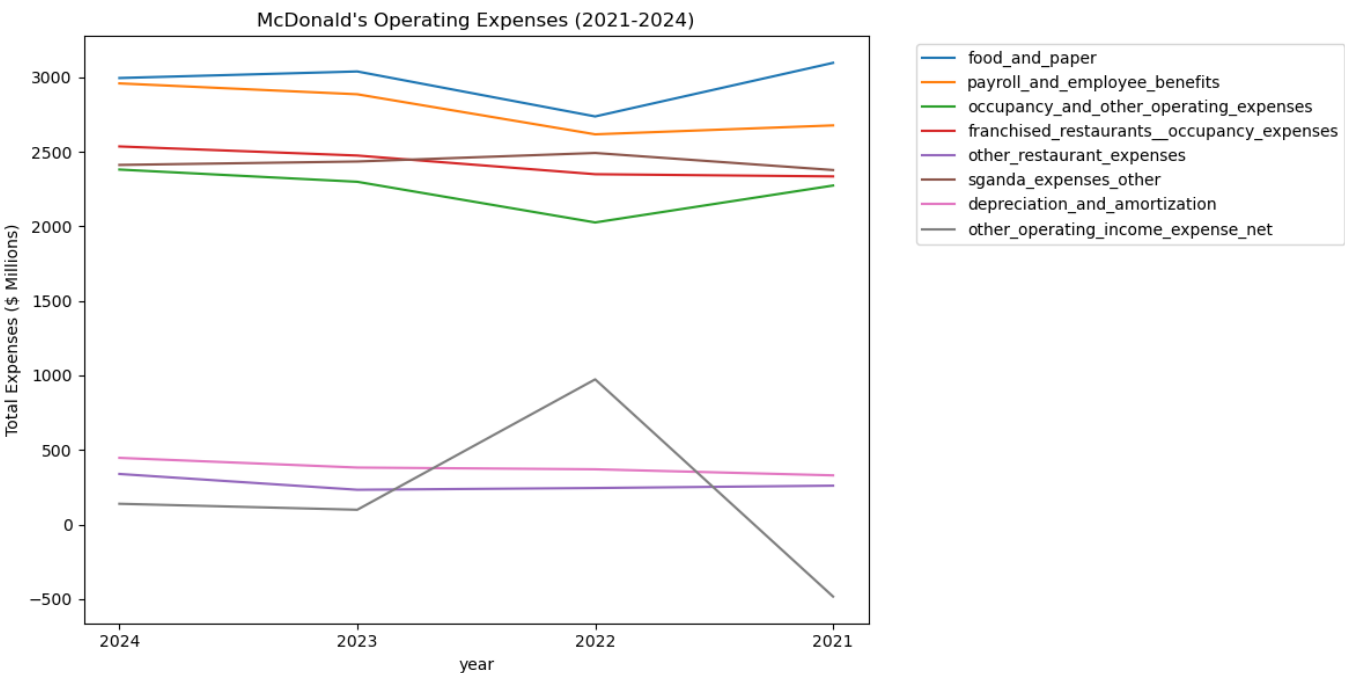


Figure 2: Operating Expense Trends

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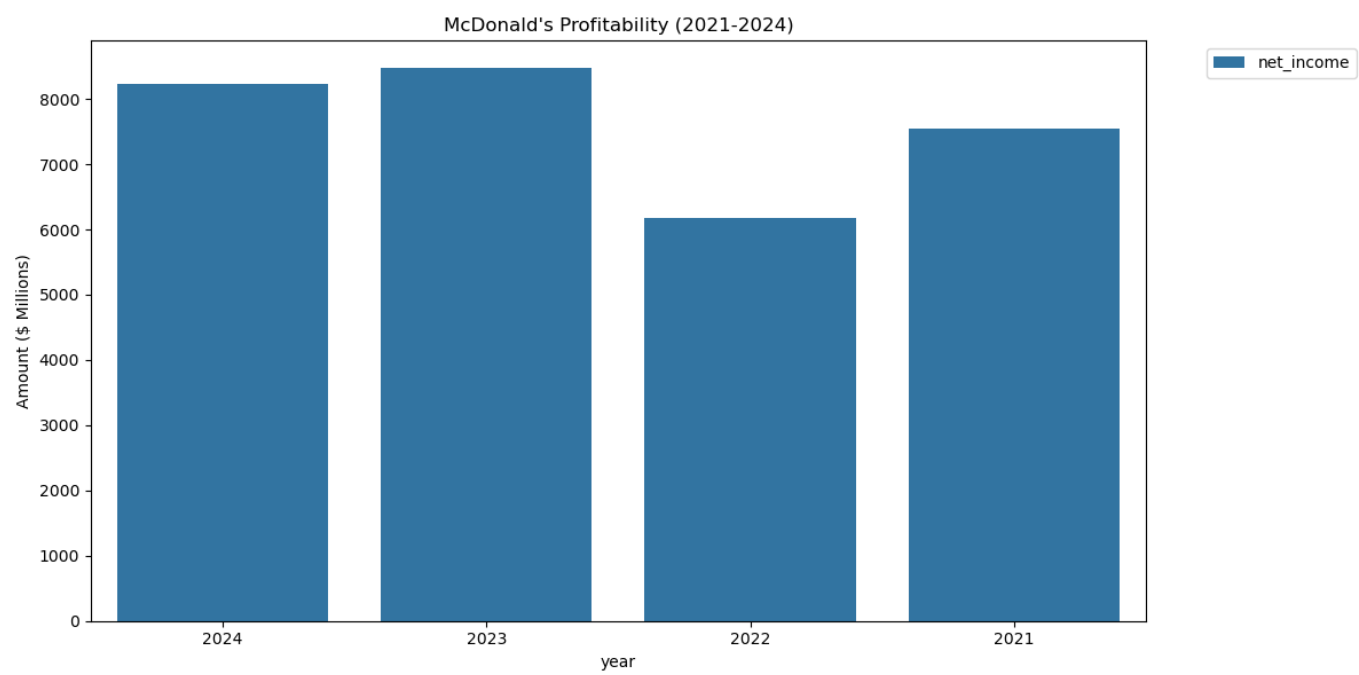


Figure 3: Profitability Metrics

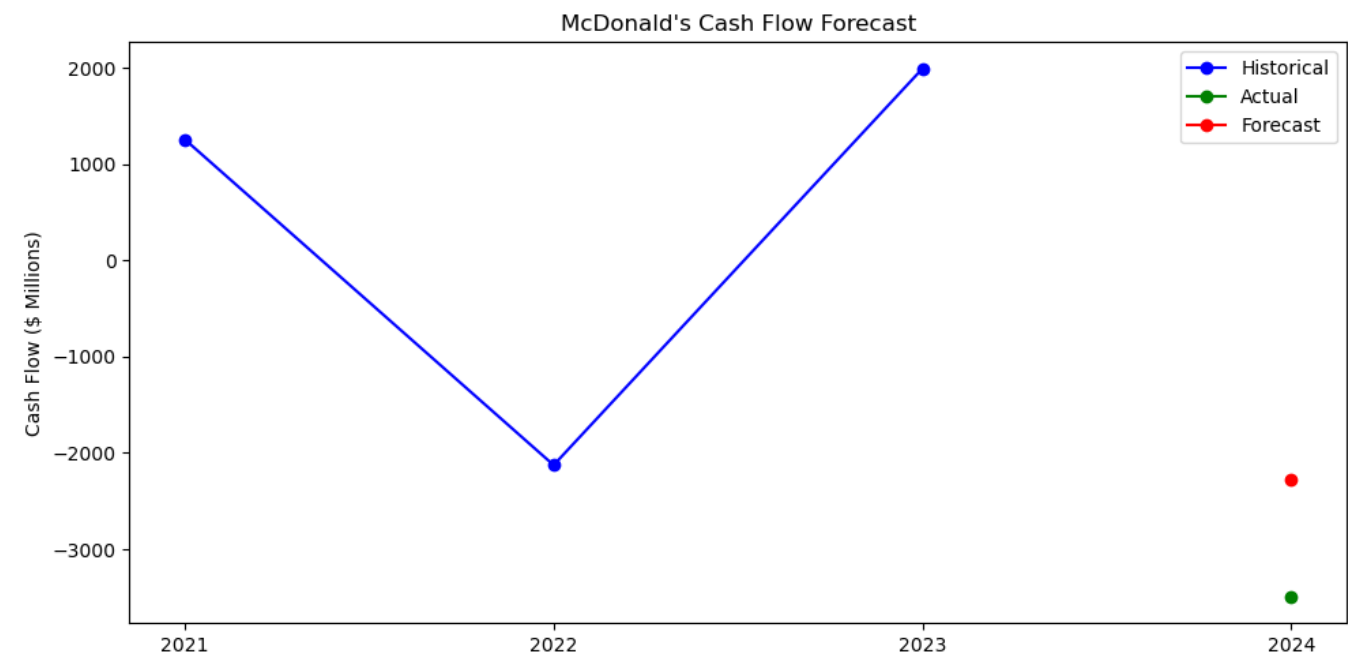


Figure 4: Cash Flow Forecast

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Discussion/Reflection

Project Assessment:

- All visualization goals were met with 3 distinct chart types
- Forecasting MAE target was achieved
- Professional report generated

Success Factors:

- Careful data cleaning and validation
- Appropriate technology choices
- Iterative development approach

Future Improvements:

- Interactive dashboard with Plotly/Dash
- More sophisticated forecasting models
- Automated report generation pipeline

Conclusion

This project confirmed my ability to:

1. Work with real financial data
2. Create meaningful visualizations
3. Implement forecasting models
4. Communicate results professionally

Next steps could include:

- Adding user authentication

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- Creating a web-based dashboard
- Incorporating real-time data feeds

References

Resources used:

- Pandas documentation
- Statsmodels ARIMA guide
- Seaborn visualization examples
- FPDF library documentation